CBSE Practice Worksheet

Class 8 | Mathematics | Difficulty: Hard | Set C

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Find HCF of $(x^2 - 4)$ and $(x^2 - 4x + 4)$
   - (A) $x - 2$
   - (B) $x + 2$
   - (C) $(x-2)^2$
   - (D) $x^2 - 4$
2. Solve: $\sqrt{(x + 7)} = x - 5$
   - (A) $x = 9$
   - (B) $x = 2$
   - (C) $x = 9$ or 2
   - (D) No solution
3. Solve: $|2x - 5| = 9$
   - (A) $x = 7$ or -2
   - (B) $x = 7$ or 2
   - (C) $x = -7$ or 2
   - (D) $x = -7$ or -2
4. Solve: 2^(x+1) = 32
   - (A) $x = 3$
   - (B) $x = 4$
   - (C) $x = 5$
   - (D) $x = 6$
5. Find: $(-1)^{1}$■■ + (-1)■■
   - (A) 0
   - (B) 2
   - (C) -2
   - (D) 1
6. What is the cube root of -125?
   - (A) 5
   - (B) -5
   - (C) ±5
   - (D) undefined
7. Factorize: $x^2 - 5x - 14$
   - (A) $(x-7)(x+2)$
   - (B) $(x+7)(x-2)$
   - (C) $(x-7)(x-2)$
   - (D) $(x+7)(x+2)$
8. Find: $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$
   - (A) 2
   - (B) 8
   - (C) 15

(D) $\sqrt{15}$

9. Simplify: $(x^3 - 8)/(x - 2)$
   - (A) $x^2 + 2x + 4$
   - (B) $x^2 - 2x + 4$
   - (C) $x^2 + 4$
   - (D) $x^2 - 4$

10. Solve: $3(x-2)/4 = (2x+1)/3$
    - (A) $x = 14$
    - (B) $x = 22$
    - (C) $x = 26$
    - (D) $x = 30$

Section B: Fill in the Blanks (1 mark each)

11. Rationalize: $1/(\sqrt{3} + 1) = (\sqrt{3} - 1)/$___

12. If $x + y = 7$ and $xy = 12$, then $x^2 + y^2 =$ ___

13. HCF × LCM of 12 and 18 = ___

14. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 -$ ___

15. $(x + 1/x)^2 - (x - 1/x)^2 =$ ___

16. $\sqrt[3]{-64} =$ ___

17. If $x^2 - 6x + 8 = 0$, then $x =$ ___ or ___

18. If 5■ = 125, then x = ___

Section C: Short Answer Questions (2 marks each)

19. If α and β are roots of $x^2 - 5x + 6 = 0$, find $\alpha^3 + \beta^3$

20. Solve: log■(x - 1) + log■(x + 1) = 3

21. Find the value of x: $3^{(2x-1)} = 27^{(x-2)}$

22. If $x = 2 + \sqrt{3}$, find $x + 1/x$

23. Factorize: $8x^3 - 27$

24. Simplify: $(2 + \sqrt{3})/(2 - \sqrt{3})$

25. Prove: $(a + b)^3 - (a - b)^3 = 2b(3a^2 + b^2)$

--- End of Worksheet ---